

Aviroop Banerjee

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SKILLS

Languages: JavaScript/TypeScript, Java, Python, C++, SQL, Go

Backend & Cloud: Node.js, Spring Boot, REST, gRPC, AWS, GCP, Docker, Kubernetes, Terraform, Apache Kafka

AI & Data Engineering: LangChain, LangGraph, RAG, LLM agents, prompt engineering, PyTorch, TensorFlow, Whisper, DuckDB, Pandas

Databases: PostgreSQL, pgvector, Elasticsearch, Redis, DynamoDB, MongoDB, Redshift, Snowflake

Observability & Tools: OpenTelemetry, Prometheus, Jenkins, GitHub Actions (CI/CD), Git, Linux

EXPERIENCE

CommerceIQ

Bengaluru, India

Software Development Engineer

July 2024 – Present

- Cut bug triage-to-PR time from **5 hrs to 20 min** across 23+ resolved tickets by building a **multi-agent system** (LangGraph) with specialized sub-agents that localize faults from a Jira ticket's stack trace via lexical + call-graph search, generate a candidate fix, and open a PR behind human approval checkpoints.
- Replaced a legacy polling pipeline with an event-driven sync layer (S3 event triggers → AWS Lambda), reducing P99 download latency **60%**; cited by the team as a contributor to **improved client retention**.
- Spearheaded the design of a JSON-driven, PostgreSQL-backed **template service** for AllyAI, enabling dashboard assembly from reusable widgets and slashing report-build time **79%** (from 24 hours to 5 hours).
- Architected a standardized '**Golden Path**' **Node.js foundation library** adopted across the microservice ecosystem, integrating OpenTelemetry and circuit-breaker patterns to mitigate cascading failures and accelerate new service deployment by **90%**.
- Engineered a distributed alert notification service in **Java (Spring Boot)** utilizing **Apache Kafka** in a fan-out architecture, dispatching events for **75K+ SKUs** across 4 daily batch runs with at-least-once delivery guarantees.

Software Development Engineer Intern

Jan 2024 – July 2024

- Built a NestJS **RBAC microservice** automating component rendering and role provisioning, cutting bulk user onboarding from 4 hours to under 10 minutes (**24× faster**) across 5+ internal teams.
- Built a RAG pipeline (LangChain + pgvector) over 500+ brand campaigns (50K ad documents), cutting answer hallucination from **22% to 14%** (an 8pp / 35% relative drop, measured on a 200-query eval set) and powering LLM-generated budget-allocation recommendations adopted by the ads team.

Airbook

Mumbai, India

Software Engineer Intern

Sep 2023 – Jan 2024

- Achieved **67% faster** data processing and minimum **40%** query-efficiency gains by integrating DuckDB and orchestrating ETL pipelines across BigQuery, Redshift, and Supabase.
- Attained **74.6% accuracy** on benchmark query execution, resulting in a **7x surge** in query frequency, by developing a **NLQ-to-SQL** system using t5-base and Meta CodeLlama-13B-Instruct.

Listed

Bengaluru, India

Backend Engineer Intern

Mar 2023 – Sep 2023

- Orchestrated a multi-modal recommendation pipeline for 11M users, leveraging OpenAI Whisper for speech-to-text extraction from video ads and BERT for semantic entity mapping to achieve **97%** accuracy in automated affiliate link generation.

PROJECTS

Generative AI & Financial Reporting Platform — Sample Report | *scikit-learn, Machine learning*

- Developed an ensemble ML pipeline (RandomForest, LightGBM, Logistic Regression) with TF-IDF feature engineering and SMOTE, processing bank transactions and achieving **92% classification accuracy**.
- Automated **scheduled reporting** and delivered goal-based spend optimization recommendations via actionable AI insights using GitHub Actions, Gemini, and Gmail API.

EDUCATION

B.Tech in Computer Science and Engineering (GPA-9.34)

Maulana Abul Kalam Azad University of Technology

Oct. 2020 – May 2024

India